How Rivers Keep Time

A river is the slowest clock most of us will ever look at. The water moves quickly enough, but the shape of the channel, the width of the valley and the depth of the canyon record changes that took thousands or millions of years. Standing on a bridge, you see a single frame of a film that started long before anyone was there to watch it.

Geologists read that film backwards. A terrace high above the present channel marks where the river once ran, before it cut down through the rock beneath it. A layer of rounded gravel on a hillside tells them that water, not ice or wind, carried those stones. Each clue is small, but together they describe a landscape that is always on its way to becoming something else.

The cutting itself is mostly done by grit. Clean water flowing over solid rock does surprisingly little. Sand and pebbles carried by the current are what grind the bed down, a few millimetres in a good year, and almost nothing in a quiet one. The work is uneven: most of it happens during floods, when the river carries more sediment in a single week than in the rest of the year combined.

That unevenness makes rivers hard to predict. A model that averages the flow over a decade will underestimate the damage a single storm can do, and overestimate what the river achieves in a dry spell. Engineers who design bridges and levees have learned this the expensive way, often more than once.

Meanders are another kind of clock. A river on a wide, flat plain rarely runs straight. Small bends grow because water on the outside of a curve moves faster and erodes the bank, while slower water on the inside drops its load and builds a bar. Over time the bends exaggerate themselves until a loop is cut off during a flood, leaving a crescent-shaped lake behind.

Those abandoned loops, called oxbow lakes, fill slowly with silt and plants. Pollen trapped in their mud gives researchers a record of the vegetation that grew nearby, layer by layer, sometimes for thousands of years. A lake that began as a river's mistake becomes one of the best archives of the surrounding countryside.

Deltas keep a different kind of record. Where a river meets the sea, it slows down and drops whatever it is carrying. The delta grows outward, splits into channels, abandons some of them and builds new lobes elsewhere. Satellite images taken a few decades apart show coastlines shifting by kilometres, which is fast by geological standards and slow by human ones.

People have always settled along rivers, and in doing so they have changed the clocks they depend on. Dams trap sediment that would have built the delta downstream. Straightened channels carry floods faster to the next town. Gravel mining lowers the bed and undermines bridges. Each change is local, but the river connects them all, so the effects travel.

None of this is a reason to treat rivers as fragile. They are enormously persistent, and many have outlasted mountains. It is a reason to read them carefully, to notice the terrace, the gravel and the oxbow, and to remember that the frame we see from the bridge is not the whole film.

The next time you cross a river, look for the evidence of its history. Notice which bank is steeper, where the sand collects, and how high the debris from the last flood was left in the trees. You are reading the same clues a geologist would, and the river is still writing.